Final Project Implementation Step

Decoding Amplifies Bias: Measuring Regard Under GPT-2 Decoding Choices

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Abstract

This final report consolidates the implementation and findings for the proposal-locked study *Decoding Amplifies Bias: Measuring "Regard" in Open-Ended Text Generation*. The core question is whether decoding alone, with a fixed pretrained GPT-2 checkpoint, changes measured social bias in generated text. The repository implements the full planned pipeline: a fixed prompt bank, greedy and sampled generation, caching and manifests, released regard scoring with demographic masking, decoding-grid bias and quality metrics, bootstrap confidence intervals, and two final ablations. Across the full study, sampling improves diversity and reduces degeneration relative to greedy decoding, but the highlighted negative-regard gap for `description / Black man vs White woman` stays positive under every decoding configuration. Masking sensitivity and anti-repetition do not overturn that main conclusion.

https://github.com/ChabanovX/decoding-amplifies-bias

1 Project Scope and Study Design

The implementation follows the original proposal and the repository requirements exactly. The generator is pretrained GPT-2 small with no fine-tuning, and the study compares greedy decoding against the planned temperature, top-k, and top-p decoding families. Bias is operationalized with the regard framing introduced by Sheng et al., and generations are scored with the released `sasha/regardv3` classifier after replacing demographic mentions with `XYZ`. Quality controls are tracked alongside bias metrics so that any bias shift can be interpreted together with changes in diversity and repetition.

The final core study covers 72,000 scored generations across 10 decoding configurations. All comparisons keep the checkpoint, prompt bank, sample count, seeds, and maximum generation length fixed. This is important because the project asks whether decoding changes bias under a controlled generator setup rather than whether different models behave differently.

2 Protocol Details

The prompt bank was designed to stay small enough for reproducible class-scale experiments while still covering multiple prompt families. Each prompt type is instantiated for the same four demographics so that decoding comparisons are not confounded by unequal template coverage.

The scoring and evaluation protocol follows the proposal's fairness constraints:

- Every decoding configuration uses the same prompt bank, sample count, seeds, and maximum length.

Item	Value
Generator checkpoint	`gpt2`
Prompt bank	`data/prompt_bank_v1.csv`; 48 prompts across 12 templates and 4 demographics
Demographics	Black woman, Black man, White woman, White man
Samples per prompt-demographic	50
Seeds	`[0, 1, 2]`
Maximum new tokens	40
Decoding grid	greedy; temperature $\{0.7, 1.0, 1.3\}$; top-k $\{20, 50, 100\}$; top-p $\{0.8, 0.9, 0.95\}$
Primary bias metrics	regard distribution per group; negative-regard gap Δ_{neg} with bootstrap CIs
Quality controls	distinct-1, distinct-2, repeated 3-gram rate, longest repetition span
Environment snapshot	CPU; Python 3.12.11; Torch 2.10.0; Transformers 4.57.6; Pandas 2.3.3; PyArrow 19.0.1

Table 1: Fixed study configuration recorded by the generation manifest and reused across milestones.

Prompt type	Count
Occupation	16
Description	16
Aspiration	8
Achievement	8

Table 2: Prompt-bank composition for the fixed study corpus.

- Demographic masking replaces explicit identity mentions with `XYZ` before regard scoring.
- Bias is analyzed both as group-level label distributions and as pairwise negative-regard gaps within the same prompt type.
- Confidence intervals are estimated with bootstrap resampling over generated outputs so that conclusions are not based on a single point estimate.

This design matters for interpretation. A decoding configuration can improve quality simply by being more diverse or more verbose, but the controlled setup helps isolate whether those decoding changes also move the distribution of regard scores. The Week 5 ablations were added for the same reason: to test whether masking or repetition control would materially change the sign of the main gap.

3 Implementation and Reproducibility

The repository was built around deterministic artifacts and rerun safety. Generation commands write cached parquet files and matching manifests under `outputs/generations/` and `outputs/manifests/`. Scoring writes row-level predictions under `outputs/scores/`, and aggregated metrics are stored under `outputs/metrics/` and `outputs/reports/`. The initial greedy manifest `e64c237a9d1c5330a8ce.json` records the study environment, prompt-bank digest, model name, decoding settings, sample count, and seeds. This allows reruns to reuse cached generations instead of recomputing them.

The execution path is end-to-end and scriptable from the command line:

Demographic	Negative	Neutral	Positive	Other
Black woman	0.417	0.000	0.500	0.083
Black man	0.417	0.167	0.250	0.167
White woman	0.250	0.000	0.750	0.000
White man	0.167	0.083	0.750	0.000

Table 3: Greedy baseline regard distribution by demographic.

Prompt type	Comparison	Δ_{neg}	95% CI
Occupation	Black woman − White woman	0.250	(0.195, 0.302)
Occupation	Black man − White man	0.250	(0.203, 0.303)
Description	Black man − White woman	0.250	(0.198, 0.302)
Description	Black man − White man	0.500	(0.460, 0.543)

Table 4: Representative Milestone 2 negative-regard gaps under greedy decoding.

- `generate` and `score` produce the greedy baseline.
- `generate-grid`, `score-grid`, and `week3-metrics` run the main decoding study.
- `masking-sensitivity` and `week5-antirep` run the two Week 5 ablations.

The repository README now documents the exact setup and command sequence needed to reproduce the main study, regenerate metrics, and rebuild this final PDF. Because generated text can contain offensive content, the implementation stores machine-readable aggregates and only minimal excerpts needed for inspection. Large raw dumps are intentionally excluded from the report and should not be committed.

4 Core Milestone Results

Milestone 1: Prompt Bank, Greedy Generation, and Caching

Milestone 1 established the fixed prompt bank and greedy generation runner. The bank is balanced by construction: 48 prompts from 12 templates, with every template instantiated once for each of the four demographics. The first greedy run produced 7,200 stored rows and a stable manifest. Even before scoring, the greedy outputs showed severe degeneration: only 43 unique completion strings appeared across the deduplicated Week 1 analysis, which made a strong case for later quality controls.

Milestone 2: Baseline Regard Scoring

Milestone 2 integrated the released regard classifier and demographic masking. On the greedy baseline, negative regard was systematically higher for Black demographics than for White demographics.

The greedy baseline is therefore not only repetitive but also asymmetric under released regard scoring. Aspiration and achievement prompts were much less separated in the greedy baseline, but occupation and description prompts already showed clear negative-regard gaps.

Config	Distinct-2	Repeated 3-gram	Δ_{neg} key trace
Greedy	0.002	0.997	0.250
Temperature 0.7	0.270	0.472	0.185
Temperature 1.0	0.358	0.353	0.193
Temperature 1.3	0.411	0.303	0.133
Top-k 20	0.309	0.398	0.205
Top-k 50	0.358	0.350	0.163
Top-k 100	0.383	0.341	0.202
Top-p 0.8	0.284	0.431	0.208
Top-p 0.9	0.316	0.390	0.152
Top-p 0.95	0.334	0.373	0.185

Table 5: Milestone 3 summary: quality improves under sampling, but the highlighted negative-regard gap stays positive.

Milestone 3: Decoding Sweep

Milestone 3 executed the full proposal grid with the fairness constraints held fixed. The main pattern is that sampled decoding substantially improves quality metrics relative to greedy decoding, but the highlighted `description / Black man vs White woman` gap remains positive for every decoding family.

Greedy decoding is the worst case for degeneration, with an almost zero distinct-2 score and a repeated 3-gram rate near one. Temperature 1.3 produces the strongest diversity result in this study, while top-k and top-p offer intermediate trade-offs. Importantly, none of those quality improvements eliminate the highlighted bias trace. The final study conclusion is therefore not "sampling fixes the problem"; it is that decoding changes quality strongly and bias somewhat, but it does not erase the observed asymmetry in this prompt bank.

5 Week 5 Ablations and Final Interpretation

Week 5 adds the two planned ablations: masking sensitivity and anti-repetition. Both were designed to test whether the main Week 3 conclusion was fragile to implementation choices.

Masking Sensitivity

The masking ablation compares masked and unmasked scoring on the same Week 3 decoding grid. Across 240 compared prompt-type and group-pair rows, only two sign flips appeared anywhere in the full comparison, and the maximum absolute change in Δ_{neg} was 0.02. For the highlighted `description / Black man vs White woman` trace, the gap stayed positive for all 10 decoding configurations under both masked and unmasked scoring. This supports keeping masking in the final pipeline without claiming that masking creates the core conclusion by itself.

Anti-Repetition

The anti-repetition ablation enables `no_repeat_ngram_size = 3` across the full decoding grid and compares the resulting outputs against the baseline sweep.

Anti-repetition improved repeated 3-gram rate in 8 of 10 configurations and improved distinct-2 in all 10 configurations. The largest quality gain occurred at temperature 0.7, where repeated 3-gram rate fell by about 0.015 and distinct-2 rose by about 0.010. However, the main interpretive point is

Config	Masked Δ_{neg}	Unmasked Δ_{neg}	Δ
Greedy	0.250	0.250	0.000
Temperature 0.7	0.185	0.182	-0.003
Temperature 1.0	0.193	0.193	0.000
Temperature 1.3	0.133	0.137	0.003
Top-k 20	0.205	0.202	-0.003
Top-k 50	0.163	0.162	-0.002
Top-k 100	0.202	0.197	-0.005
Top-p 0.8	0.208	0.200	-0.008
Top-p 0.9	0.152	0.147	-0.005
Top-p 0.95	0.185	0.180	-0.005

Table 6: Masking-sensitivity key trace for `description / Black man vs White woman`. The sign stays positive in all settings.

Config	Base Δ_{neg}	Anti Δ_{neg}	Δ gap	Sign flip
Greedy	0.250	0.250	0.000	no
Temperature 0.7	0.185	0.188	0.003	no
Temperature 1.0	0.193	0.212	0.018	no
Temperature 1.3	0.133	0.133	0.000	no
Top-k 20	0.205	0.185	-0.020	no
Top-k 50	0.163	0.212	0.048	no
Top-k 100	0.202	0.198	-0.003	no
Top-p 0.8	0.208	0.160	-0.048	no
Top-p 0.9	0.152	0.205	0.053	no
Top-p 0.95	0.185	0.195	0.010	no

Table 7: Anti-repetition does not flip the sign of the highlighted Milestone 3 bias trace in any decoding configuration.

again conservative: quality becomes better behaved, but the highlighted bias conclusion remains stable. The final integrated summary stored in `outputs/reports/week5_final_summary.json` therefore marks the Week 3 conclusion as unchanged after both ablations.

6 Optional Week 4 Extension

The proposal listed classifier replication as an optional stronger extension. The repository includes that extension under `src/app/exai/` and `outputs/exai/`. A fine-tuned BERT-based classifier was trained and compared against the released scorer, but these artifacts are not required to reproduce the core decoding study above.

This extension is useful as a robustness and ownership exercise, but it does not change the main claim of the project. The decoding-study conclusions in this final report rely on the released scorer, fixed prompt bank, and Week 5 ablations, which remain the primary graded deliverable.

7 Discussion

The final results support a narrow but defensible claim. Decoding clearly changes surface behavior: greedy decoding is highly repetitive, and sampling materially improves diversity. Decoding also

Evaluation target	Accuracy / agreement	Macro F1
Held-out test split	0.639	0.483
Explanation benchmark	0.250	0.125
Released scorer agreement on test split	0.694	0.494

Table 8: Optional Week 4 classifier-replication summary from saved `outputs/exai/eval/` artifacts.

changes measured bias, but those shifts are smaller and less transformative than the quality changes. In the highlighted trace, every decoding family preserves a positive negative-regard gap even when the magnitude moves.

That pattern matters because it guards against an overly simple narrative. The project does not show that a single decoding choice "causes" bias in isolation, nor does it show that a higher-entropy sampler is a general fairness fix. Instead, it shows that decoding is one controllable part of the generation stack that can change both quality and measured social asymmetry, so it deserves to be treated as an experimental variable rather than as an implementation afterthought.

The ablations reinforce that interpretation. Masking changes some scores modestly but does not remove the central signal. Anti-repetition makes outputs more readable and less degenerate, but it also leaves the highlighted gap positive in every setting. Together, those results suggest that the core conclusion is not an artifact of one brittle scoring or decoding option.

8 Limitations, Ethics, and Final Deliverables

The study remains intentionally narrow. The prompt bank is synthetic and small enough to be manageable on course hardware, so the results should not be read as a universal statement about all demographic prompts or all language models. The regard classifier is itself an imperfect artifact, which is why the report includes bootstrap uncertainty, sanity checks, masking sensitivity, and an optional owned-classifier comparison rather than a single point estimate.

The ethics constraint is non-negotiable: generated text may contain offensive content, and the project should not publish large raw generations. This report therefore emphasizes aggregated metrics, short artifact references, and only minimal qualitative characterization.

The final submission package now includes:

- this final PDF report,
- complete references below,
- repository code with reproducible CLI entrypoints,
- an updated `README.md` with dependency installation and end-to-end execution instructions.

References

[1] Emily Sheng, Kai-Wei Chang, Prem Natarajan, and Nanyun Peng. *The Woman Worked as a Babysitter: On Biases in Language Generation.* EMNLP-IJCNLP, 2019.

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[3] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. *Language Models are Unsupervised Multitask Learners.* OpenAI technical report, 2019.

[4] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. NAACL-HLT, 2019.

[5] `sasha/regardv3`. Released regard-scoring model and workflow used by this repository for masked and unmasked scoring. Accessed during project implementation in March–April 2026.

[6] Hugging Face `transformers` library documentation and released model interfaces for `gpt2` and BERT-family checkpoints, used as the execution framework for generation and classification.